

1.7 Sums of Subspaces

In the previous sections, we studied intersections and unions of subspaces. While the intersection of two subspaces is always a subspace, the union typically is not. This raises a natural question: what is the “right” way to combine two subspaces into a larger one?

The answer is the **sum** of subspaces. Given two subspaces W_1 and W_2 , their sum $W_1 + W_2$ consists of all vectors that can be written as $\mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. This construction always yields a subspace—in fact, it is the smallest subspace containing both W_1 and W_2 .

A particularly nice situation occurs when W_1 and W_2 have only the zero vector in common. In this case, their sum is called a **direct sum**, and every vector in the sum can be decomposed uniquely into a piece from W_1 and a piece from W_2 . Direct sums are fundamental in decomposing vector spaces into simpler, independent components.

1.7.1 Definition and Properties

Definition 1.7.1 — Sum of Subspaces

Let W_1, W_2 be subspaces of a vector space V . Then we define

$$W_1 + W_2 := \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2\}.$$

Theorem 1.7.1 — Sum of Subspaces Theorem

$W_1 + W_2$ is a subspace of V . Furthermore, this is the smallest subspace containing both W_1 and W_2 .

Proof. To prove that $W_1 + W_2$ is a subspace V . We prove that three conditions of the subspace test:

- Non-emptiness: As W_1, W_2 are subspaces, they both contains $\mathbf{0}$. And since $\mathbf{0} = \mathbf{0}_{W_1} + \mathbf{0}_{W_2}$, therefore $W_1 + W_2$ is non-empty.
- Closed under addition: Let $\mathbf{a}, \mathbf{b} \in W_1 + W_2$. Then we have

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2, & \text{for some } \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2 \\ \mathbf{b} = \mathbf{w}'_1 + \mathbf{w}'_2, & \text{for some } \mathbf{w}'_1 \in W_1, \mathbf{w}'_2 \in W_2 \end{cases}$$

Then $\mathbf{a} + \mathbf{b} = \underbrace{(\mathbf{w}_1 + \mathbf{w}'_1)}_{\in W_1} + \underbrace{(\mathbf{w}_2 + \mathbf{w}'_2)}_{\in W_2} \in W_1 + W_2$. So $W_1 + W_2$ is closed under addition.

- Closed under scalar multiplication: Let $\mathbf{a} \in W_1 + W_2$, and $\lambda \in F$. Then we have $\mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Then $\lambda \mathbf{a} = \underbrace{\lambda \mathbf{w}_1}_{\in W_1} + \underbrace{\lambda \mathbf{w}_2}_{\in W_2} \in W_1 + W_2$. So

$W_1 + W_2$ is closed under scalar multiplication.

This concludes that $W_1 + W_2$ is a subspace.

To prove that $W_1 + W_2$ is the smallest subspace containing both W_1 and W_2 , we first prove that it actually contains W_1 and W_2 , then we prove that it is the smallest subspace of V containing so.

- **Containment:** Since W_1 and W_2 are subspaces, they both contain the zero vector $\mathbf{0}$. Since for any $\mathbf{w}_1 \in W_1$, we can write $\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0}$, by $\mathbf{0} \in W_2$, we have $\mathbf{w}_1 \in W_1 + W_2$. Thus $W_1 \subseteq W_1 + W_2$. The same logic applies to W_2 therefore $W_2 \subseteq W_1 + W_2$.
- **Minimality:** Let U be any subspace of V such that $W_1 \subseteq U$ and $W_2 \subseteq U$. We want to show that $W_1 + W_2 \subseteq U$. Take any element $W_1 + W_2$. By definition $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. Because $W_1 \subseteq U$, then $\mathbf{w}_1 \in U$. Similarly, because $W_2 \subseteq U$, then $\mathbf{w}_2 \in U$. At last, since U is a subspace, it is closed under addition. Therefore

$$\mathbf{w}_1 + \mathbf{w}_2 \in U \implies \mathbf{v} \in U.$$

This concludes that $W_1 + W_2$ is the smallest subspace of V containing W_1 and W_2 . ■

1.7.2 Dimension of $W_1 + W_2$

Theorem 1.7.2 — Dimension Formula of Subspace Sum

Let V be a finite-dimensional vector space, and let W_1, W_2 subspaces of V . Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof. We use the above diagram to build intuition for the proof.

Let $n = \dim(W_1 \cap W_2)$. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the intersection $W_1 \cap W_2$.

Since B is a linearly independent subset of both W_1 and W_2 , by the Basis Extension Theorem (@thm-basis-extension), we can extend it to form bases for both subspaces:

1. **For W_1 :** Extend B with $\{\mathbf{s}_1, \dots, \mathbf{s}_p\}$ such that $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 . Note that $\dim(W_1) = n + p$.
2. **For W_2 :** Extend B with $\{\mathbf{t}_1, \dots, \mathbf{t}_q\}$ such that $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 . Note that $\dim(W_2) = n + q$.

Claim. The set $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of $W_1 + W_2$.

Proof.

- S spans $W_1 + W_2$

Let $\mathbf{w} \in W_1 + W_2$. By definition, $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Since B_1 spans W_1 and B_2 spans W_2 , both $\mathbf{w}_1$ and $\mathbf{w}_2$ can be written as linear combinations of vectors in S . Thus, their sum $\mathbf{w}$ is also in $\text{span}(S)$.

- S is Linearly Independent

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0} \quad (1.7.1)$$

Rearrange the equation to isolate the terms involving $\mathbf{t}_k$:

$$\mathbf{v} := \sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = - \sum_{k=1}^q \gamma_k \mathbf{t}_k$$

Observe the vector $\mathbf{v}$. From the LHS, we have $\mathbf{v} \in W_1$. From the RHS, we have $\mathbf{v} \in W_2$. Therefore, $\mathbf{v} \in W_1 \cap W_2$.

Since $\mathbf{v} \in W_1 \cap W_2$, it can be written uniquely as a linear combination of the basis B of the intersection:

$$\mathbf{v} = \sum_{i=1}^n \delta_i \mathbf{b}_i$$

Equating this with the RHS expression of $\mathbf{v}$:

$$\sum_{i=1}^n \delta_i \mathbf{b}_i = - \sum_{k=1}^q \gamma_k \mathbf{t}_k \implies \sum_{i=1}^n \delta_i \mathbf{b}_i + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0}.$$

The set $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 and is therefore linearly independent. This forces all coefficients to be zero. Specifically, all $\gamma_k = 0$.

Substitute $\gamma_k = 0$ back into 1.7.2:

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = \mathbf{0}$$

Since $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 , it is linearly independent. This forces all $\alpha_i = 0$ and all $\beta_j = 0$.

Since all α, β, γ are zero, S is linearly independent. ■

Use dimension counting, since S is a basis for $W_1 + W_2$:

$$\begin{aligned}\dim(W_1 + W_2) &= |S| \\ &= n + p + q \\ &= (n + p) + (n + q) - n \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).\end{aligned}$$

■

Remark. **Counter-example for 3 Subspaces**

It is a coincidence that this formula resembles the “Inclusion-Exclusion Principle” from set theory. This intuition fails for three or more subspaces. The formula:

$$\dim(W_1 + W_2 + W_3) \neq \sum \dim(W_i) - \sum \dim(W_i \cap W_j) + \dim(W_1 \cap W_2 \cap W_3)$$

Counter-example: Consider three distinct lines through the origin in $\mathbb{R}^2$ (e.g., x-axis, y-axis, and $y = x$).

- LHS: The sum of three lines fills the plane, so dimension is **2**.
- RHS: $1 + 1 + 1 - (0 + 0 + 0) + 0 = 3$.

Since $2 \neq 3$, the formula is false.

1.7.3 Direct Sum of Two Subspaces

Definition 1.7.2 — Direct Sum of Two Subspaces

Let W_1, W_2 be subspaces of a vector space V . If $W_1 \cap W_2 = \{\mathbf{0}\}$, then we write $W_1 + W_2$ as

$$W_1 \oplus W_2$$

and call it the **direct sum** of W_1 and W_2 .

Example.

Consider the vector space $V = M_n(\mathbb{C})$, and two of its subspaces:

- $W_1 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = \mathbf{A}\}$ (the subspace of symmetric matrices);
- $W_2 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = -\mathbf{A}\}$ (the subspace of skew-symmetric matrices).

We can derive that $V = W_1 \oplus W_2$.

Proof. We first show that $V = W_1 + W_2$. That is, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix. Let $\mathbf{A} \in M_n(\mathbb{C})$, notice that we have the identity:

$$\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$$

where the first term is symmetric and the second term is skew-symmetric. Therefore, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix.

To establish that $W_1 \cap W_2 = \{\mathbf{O}\}$, let $\mathbf{A} \in W_1 \cap W_2$. By definition, this implies $\mathbf{A} \in W_1$ and $\mathbf{A} \in W_2$. Consequently, the identities $\mathbf{A} = \mathbf{A}^\top$ and $\mathbf{A} = -\mathbf{A}^\top$ must hold simultaneously. Substituting the former into the latter yields $\mathbf{A} = -\mathbf{A}$. This implies $2\mathbf{A} = \mathbf{O}$, and thus $\mathbf{A} = \mathbf{O}$.